Department of Electrical And Electronic Engineering
University of Dhaka
Batch - 103, EEE-58

2^{nd} Year 1^{st} Semester

2-1 All Incourse Questions

Note by
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Session: 2023-2024 (EEE-58)
Class Roll: SH-042
Reg No: 2023-916-278

Department of Electrical And Electronic Engineering, University of Dhaka

2^{nd} Year 1^{st} Semester B.Sc. Engineering Incourse Examination-2024 (EEE-57)

EEE-2101: Analog Electronics I (3 credits)

Time: 1 hour **Full Marks: 20**

Answer any three(3) of the following four(4) questions

1. Two identical pn junctions are placed in series (3+3=6)
   (a) Prove that the combination can be viewed as a single two-terminal device having and exponential characteristics.
   (b) For a tenfold change in the current, how much voltage change does such a device require.

2. For full-wave rectifier of Fig. below, i) Calculate the peak output current, and the PIV rating of the diodes. ii) Draw the input waveforms to the rectifier and the output waveforms. For both cases assume Si diodes with $V = 0.7V$. The transformer steps down a 220V,50-Hz line voltage. (3+3)=6

3. Consider the circuit shown in Fig. below, where $I_{S.Q1} = 5 \times 10^{-17}$A and $V_{BE} = 800$mV. Assume $\beta = 100$ (a) Draw the large signal model of the bipolar junction transistor (b) Determine the transistor terminal currents and voltages and verify that the device indeed operates in the active mode. (c) Determine the maximum value of R_C that permits operation in the active mode. (2+3+3) = 8

4. A student, familiar with bipolar devices, constructs the circuit shown in the figure

Department of Electrical And Electronic Engineering, University of Dhaka
2^{nd} **Year** 1^{st} **Semester B.Sc. Engineering Incourse Examination-2024 (EEE-57)**
EEE 2103 - Digital Electronics (3 credits)
Time: 1 hour **Full Marks: 25**

Answer all the following questions

1. For the following expressions, construct the corresponding logic circuit, using AND, OR gates and INVERTERs (do not simplify). (3)
   (a) $(\overline{\overline{x}\,\overline{y} + z}) + z + xy + wz$
   (b) $y = (\overline{u \oplus y}) + \overline{x}yz$
2. simplify the following expressions: (3)
$$\overline{x}_1\overline{x}_2x_3 + \overline{x}_1x_2\overline{x}_3 + \overline{x}_1x_2x_3 + x_1\overline{x}_2\overline{x}_3 + x_1\overline{x}_2x_3 + x_1x_2\overline{x}_3 + x_1x_2x_3$$
3. Draw the logic circuit implementation of a 2-bit by 2-bit multiplier circuit. (3.5)
4. Why a correction is required for BCD adders? Is this correction also necessary for binary addition? (4)
5. Draw a cascaded 8-bit comparator using two 74HC85 ICs. (2+2)
   Describe the operation of the eight-bit comparison arrangement using the following case:
   $A_7A_6A_5A_4A_3A_2A_1A_0 = 10101111$;
   $B_7B_6B_5B_4B_3B_2B_1B_0 = 101011001$
6. How can we combine two multiplexer ICs of 74HC15 to form a 16-input multiplexer? (4)
7. Draw the output response of the following J-K Flip-Flop. (3.5)
   Based on the symbol, what kind of clock pulse is given here?

Department of Electrical And Electronic Engineering, University of Dhaka
2^{nd} Year 1^{st} Semester B.Sc. Engineering Incourse Examination-2024 (EEE-57)
EEE 2103 - Digital Electronics (3 credits)

Time: 1 hour **Full Marks: 25**

Answer all the following questions

1. (a) Write the next four numbers in this hex counting sequence: E9A, E9B,E9C,E9D, ……, …… , …… , ……. (3)

   (b) What range of decimal values can be represented by a four-digit hex number?

2. Design a logic circuit whose output is HIGH only when a majority of inputs A, B and C are LOW. (4)

3. Express a half-adder in **product of sum** form. (3)

4. Why a correction is required for BCD adders? Is this correction also necessary for binary addition? (4)

5. Draw a cascaded 8-bit comparator using two 74HC85 ICs. (2+2)
   Describe the operation of the eight-bit comparison arrangement using the following case:
   $A_7A_6A_5A_4A_3A_2A_1A_0 = 10101111;$
   $B_7B_6B_5B_4B_3B_2B_1B_0 = 101011001$

6. The following figure shows the pin diagram of a 74HC147 decimal-to-BCD priority encoder. If LOW input levels appear on pins, 1,4,and 13 of the 74HC147, what will be the state of the four outputs $(A_3, A_2, A_1 and A_0)$? (All other inputs are HIGH.)

7. The following table shows the different output states of the LED segments of a BDC-to-7-Segment Decoder. Derive the expressions for ANY TWO segments, (a to g) using **k-map**, and show the circuit diagram for those three segments.

Decimal Digit	Input lines				Output lines							Display pattern
	A	B	C	D	a	b	c	d	e	f	g	
0	0	0	0	0	1	1	1	1	1	1	0	0
1	0	0	0	1	0	1	1	0	0	0	0	1
2	0	0	1	0	1	1	0	1	1	0	1	2
3	0	0	1	1	1	1	1	1	0	0	1	3
4	0	1	0	0	0	1	1	0	0	1	1	4
5	0	1	0	1	1	0	1	1	0	1	1	5
6	0	1	1	0	1	0	1	1	1	1	1	6
7	0	1	1	1	1	1	1	0	0	0	0	7
8	1	0	0	0	1	1	1	1	1	1	1	8
9	1	0	0	1	1	1	1	1	0	1	1	9

Department of Electrical And Electronic Engineering, University of Dhaka
2^{nd} Year 1^{st} Semester B.Sc. Engineering Incourse Examination-2024 (EEE-57)
EEE-2105: Solid State Physics(3 credits)

Time: 1 hour **Full Marks: 20**

Answer all following questions

1. How does Van Dar Wall force arise? Derive an expression to show the dependence of potential energy between dipoles with distance between them. (6)
2. What rules do the electrons follow to fill orbitals of an atom? (4)
3. How is the binding energy of outer electrons of Cl? Discuss in terms of effective nuclear charge. Atomic number of Cl is 17. (6)
4. Define lattice, unit cell and primitive unit cell. (4)

Department of Electrical And Electronic Engineering, University of Dhaka
2^{nd} Year 1^{st} Semester B.Sc. Engineering Incourse Examination-2024 (EEE-57)
EEE 2107 - Electromechanical Energy Conversion I

Time: 1 hour **Full Marks: 20**

Answer any three of the following questions $6\frac{2}{3} \times 3 = 20$

1. Why does the rotor of a DC motor rotate? What is back e.m.f. and what is the significance of back e.m.f.? $[3 + 3\frac{2}{3}]$

2. Derive the condition for maximum efficiency of a DC motor.

   A 4 pole, 32 conductor, lap-wound DC shunt generator with terminal voltage of 200 volts delivering $12\,A$ to the load has $r_a = 2$ and field circuit resistance of 200Ω. It is driven at 1000 r.p.m. Calculate the flux per pole in the machine. If the machine has to be run as a motor with the same terminal voltage and drawing $5\,A$from the mains, maintaining the same magnetic field, find the speed of the machine. $[2\frac{2}{3} + 4]$

3. What is armature torque? Prove that the armature torque T_a is directly proportional to ΦI_a. Why shaft torque is less than armature torque? $[1 + 4 + 1\frac{2}{3}]$

4. Draw the T_a/I_a characteristics of a DC motor and explain.

   A belt-driven 1000 kW, shunt generator running at 300 r.p.m. on 220 V bus-bars continues to run as a motor when the belt breaks, then taking 10 kW. What will be its speed? Given armature resistance $= 0.025\Omega$, field resistance $= 60\Omega$ and contact drop for each brush $= 1V$, Ignore armature resistance. $[3 + 3\frac{2}{3}]$

Department of Electrical And Electronic Engineering, University of Dhaka
2^{nd} Year 1^{st} Semester B.Sc. Engineering Incourse Examination-2024 (EEE-57)
EEE 2107 - Electromechanical Energy Conversion I

Time: 1 hour **Full Marks: 20**

Answer any three of the following questions $6\frac{2}{3} \times 3 = 20$

1. Draw the circuit diagram of a single-loop generator and explain how a sinusoidal voltage is generated in this generator. Draw the waveform if the rotor starts to rotate 30° apart from the reference point. $[1 + 4 + 1\frac{2}{3}]$

2. Describe the construction of Pole-Cores and Pole-Shoes. How laminations are made self-locking? Derive the EMF equation of a generator. $[1\frac{2}{3} + 1 + 4]$

3. Explain the iron loss of a generator.
   A 4-pole lap-winding D.C. shunt generator has a useful flux per pole of 0.07 Wb. The armature winding consists of 220 turns of 0.004Ω resistance. Calculate the terminal voltage when running at 900 r.p.m. if the armature current is 50A. $[2 + 4\frac{2}{3}]$

4. Explain different efficiencies used to evaluate the performance of a DC generator. Derive the condition for maximum efficiency. What are the components of constant losses? $[2 + 3 + 1\frac{2}{3}]$

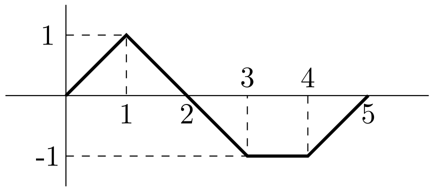
5. Is there any effect on the flux of a generator when external load is connected? What is it called? Explain with suitable diagram how this effect happens. $[1 + 1 + 4\frac{2}{3}]$

Department of Electrical And Electronic Engineering, University of Dhaka
2^{nd} Year 1^{st} Semester B.Sc. Engineering Incourse Examination-2024 (EEE-57)
EEE-2109: Signals and Systems

Time: 1 hour **Full Marks: 25**

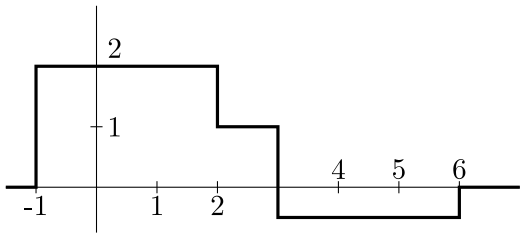
Answer any five of the following questions

1. What do you understand by signals and system? Consider the following system $x(t)$ and sketch the signal, $y(t) = -1.3x(t) + 1$. (5)



2. (a) Consider the signal $x(t)$ shown in the following figure. Sketch the signal $y(t)$. (5)

$$y(t) = x(-4t + 2)$$



(b) Consider a continuous-time exponential signal, $x(t)$. If the signal is periodic then calculate its fundamental frequency.

$$x(t) = Ce^{at}, \text{where } C \text{ is real and } a = j\omega_0$$

3. Define energy and power of signal. Calculate power and energy of the following signal (5)

$$x(t) = \begin{cases} 5\cos(\pi t) & -1 \le t \le 1 \\ 0 & \text{Otherwise} \end{cases}$$

4. Determine whether the signal is periodic or not, and if so, find the fundamental period of the signal (5)

$$y(t) = \cos(2t) + \sin(3t)$$

5. Write the names of elementary signals. Define the unit impulse function and explain its sampling and shifting properties. (5)

6. Estimate and draw the double-sided amplitude and phase spectrum of the following signal (5)

$$x(t) = 2\cos(10\pi t + \frac{\pi}{4}) + 4\sin(30\pi t - \frac{\pi}{6})$$

7. Convert the composite CT signal to DT signal and calculate its fundamental period N. (5)

$$x(t) = 3\cos(1.5\omega t) + 5\cos(5\omega t) + 8\cos(10\omega t) + \cos(20\omega t)$$

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2^{nd} Year 1^{st} Semester B.Sc. Engineering Incourse Examination-2024 (EEE-57)
EEE-2109: Signals and Systems

Time: 1 hour **Full Marks: 25**

Answer all the following questions

1. What is convolution? Explain the process with proper diagram. (5)

2. Consider the following two sequences $x[n]$ and $h[n]$ and calculate convolution of the sequences. (5)

$$x[n] = \{1, \underset{\uparrow}{0}, 0.5, 0.25\} \qquad h[n] = \{0.5, \underset{\uparrow}{1}, 0.6\}$$

3. Consider the two sequences $x[n]$ and $h[n]$ for a positive value of $\alpha > 1$. Calculate convolution of the two signals (8)

$$x[n] = \begin{cases} 1, & 0 \le n \le 4 \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad h[n] = \begin{cases} a^n, & 0 \le n \le 6 \\ 0, & \text{otherwise} \end{cases}$$

4. Calculate the convolution of the following two signals (7)

$$x[n] = \begin{cases} 1, & 0 < t < T \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad h[n] = \begin{cases} t, & 0 < t < 2T \\ 0, & \text{otherwise} \end{cases}$$

Department of Electrical And Electronic Engineering, University of Dhaka
2^{nd} Year 1^{st} Semester B.Sc. Engineering Incourse Examination-2024 (EEE-57)
MAT 2101 - Ordinary and Partial Differential Equations
Time: 1 hour **Full Marks: 20**

Answer any four of the following questions

1. Define differential equation, partial differential equation, linear differential equation, order and degree of a differential equation with example. (5)
2. When the differential equation $M(x,y)dx + N(x,y)dy = 0$ said to be exact? Solve the differential equation $(x^2 + y^2 + 1)dx - 2xydy = 0$. (5)
3. Define homogeneous differential equation. Solve the differential equation (5)
$$\frac{dy}{dx} = \frac{x + 2y - 3}{2x + y - 3}$$
4. State the existence and uniqueness theorem. Does a unique solution of the IVP $y' = 3y^{\frac{2}{3}}, y(1) = 0$ exist? If the solution exists, then find the solution. (5)
5. Define integrating factor (I.F). Solve the differential equation $y \ln y dx + (x - \ln y)dy = 0$ (5)